

Research paper

Security Information Event Management data acquisition and analysis methods with machine learning principles

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ABSTRACT

In the face of increasing global disruptions, the cybersecurity field is confronting rising threats posed by offensive groups and individual hackers. Traditional security measures often fall short in detecting and mitigating these sophisticated attacks, necessitating advanced intrusion detection methods. The goal of our study is to develop robust network intrusion detection methods using machine learning techniques. In addition, we evaluate the effectiveness of various machine learning models in detecting network intrusions. Model performances are optimized through hyperparameter tuning and feature selection. A range of classification and clustering models have been employed. Data from SIEM systems capturing real-time statistics from cloud-hosted Windows virtual machines has been gathered and augmented with web attack logs from CICIDS2017, each comprising approximately fifteen thousand rows. Hyperparameter tuning, data normalization, standardization and feature selection techniques for model optimization have been used in our study. The research showcases the potential of machine learning in enhancing network intrusion detection capabilities. The findings underscore the effectiveness of the Random Forest Classifier (0.97) and highlight the importance of utilizing diverse datasets and advanced optimization techniques. This study offers valuable insights and sets a foundation for future advancements in cybersecurity strategies and intrusion detection systems.

1. Introduction

In recent years, the global landscape has witnessed a notable surge in cybersecurity threats that pose risks to industries and the systems and servers of companies worldwide. This surge is chiefly driven by the proliferation of cloud-based devices and the abundance of sensitive online data [1]. In response, cybercriminals are continuously refining their tactics, employing increasingly sophisticated methods to evade detection and enhance their success rates. These malevolent actors are ceaselessly probing for system vulnerabilities and weaknesses that can be exploited [2]. As reported by Kaspersky [3], a concerning 41.6% of computers within the energy sector exhibit signs of infection by mal-

ware or network intrusion samples, including crypto miners, worms, spyware and traffic flooders, among others [4]. Such infections pose a tangible threat to the availability of local networks and, more critically, raise the specter of data breaches and loss of control over vital devices.

To address these formidable challenges, security analysts must proactively implement a robust cybersecurity strategy [5]. This approach encompasses a range of measures, including employee training, adherence to best practices, personal vigilance, and the deployment of security programs and technologies [2]. Therefore, it becomes paramount to adhere to best practices in information and communications technology (ICT) hygiene. This entails practices such as employing multi-factor authentication and utilizing strong, resilient pass-

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words, exercising caution when encountering suspicious links or emails, and maintaining up-to-date official software. However, many mid-sized businesses grapple with fragmented intrusion prevention tools, often employing numerous disparate instruments that lack interoperability [6]. What is needed is a unified platform capable of monitoring, controlling, and managing security devices. Enter Security Information Event Management (SIEM), a novel security tool designed to address this challenge. SIEM serves as a data consolidation point, aggregating information from diverse sources and collating event occurrences from various layers of network traffic and detection programs, including intrusion detection systems (IDS), firewalls, anti-malware software, and more [7,8]. It is imperative to apply data cleansing, normalization, and stabilization techniques to mitigate common noise and incompleteness issues when assembling data into a comprehensive big data cluster [9].

SIEM functions as a passive hub during the collection and analysis of inbound data, whereas firewalls proactively intercept and filter network traffic from potential intruders [6]. SIEM can also be deployed on popular cloud platforms like Amazon Web Services (AWS), Microsoft Azure, and Google Cloud, offering enhanced maintenance flexibility and accessibility [10]. It is worth noting that while cloud services provide on-demand access to information technology (IT) infrastructure and resources, they also present unique challenges, including elevated costs and implementation complexities.

In response to these challenges, Microsoft has streamlined the integration process by collaborating with SIEM partners such as IBM QRadar, Splunk, and ArcSight within the Azure Sentinel VM environment [11]. Azure Sentinel VM operates seamlessly with cloud infrastructure and establishes connections with security data sources and firewalls [12]. Additionally, various leading companies offer individual SIEM software packages, making them available to organizations and online users in the market. A notable example is Cyberguard's data loss prevention (DLP) software, exemplified by its use in Kazakhstan's Kazdream, where it effectively safeguards against the loss of confidential information, identifies potential risks, and streamlines employee productivity monitoring [13].

In an age defined by the pervasive presence of digital technology and relentless technological progress, the ever-evolving threat landscape of cyberattacks presents a formidable challenge to organizations worldwide [14,15]. To effectively navigate this dynamic landscape, cybersecurity strategies must remain fluid and continually embrace innovation. Within this evolving context, Security Information Event Management (SIEM) systems have not only assumed the role of safeguarding digital assets but have also become catalysts for proactive, data-driven cybersecurity [16].

Initially conceived as tools for log aggregation and correlation, SIEM systems have undergone a profound transformation, emerging as intelligent and multi-faceted platforms. Among the most innovative aspects explored in this study is the integration of Artificial Intelligence (AI) and Machine Learning (ML) into SIEM systems [16]. This infusion of cognitive capabilities empowers SIEM platforms to meticulously analyze vast datasets with a level of precision akin to human cognition. These systems exhibit the ability to learn from historical data and dynamically adapt to emerging threats in real-time. In this exploration, we delve deep into the methodologies and practical applications of AI/ML within the SIEM domain, shedding light on how these transformative technologies redefine the very foundations of detection, correlation, and response mechanisms in the realm of cybersecurity.

The rest of the paper is organized as follows; some exited models based on ML will be illustrated in the second section. Then, the study mythology will be explained in the third section. This section allows readers to understand the process of our study and experiment. Results of evaluation will be shown in the fourth section with their discussion. Finally, a section of conclusion and some future works will finish the paper.

2. Literature review

In the realm of cybersecurity, SIEM systems have emerged as pivotal instruments for organizations seeking to defend their digital assets and data. As technology evolves and threats become increasingly sophisticated, the need for a comprehensive and adaptive security solution has never been more critical. This literature review section will present some works related to SIEM context.

2.1. Machine learning models

Around the world, modern scientists research the advantages of AI in the area of information security. There are an increasing number of studies devoted to pointing analyses of aspects of this sphere's application. For instance, [17] investigate implementation problems of IDS or SIEM systems using ML approaches to detect malicious penetration for the local web of internet of things devices (IoT) [18,19]. Authors consider that there needs to be a multilayered IDS system for IoT devices because of the range of possible inconsistent behaviors of gadgets and risks of data leakage. Accessories of this kind could hold various sensitive data that criminals wish to steal by attacking attractive targets. The paper proposes a supervised classification of traffic behavior for each network member. First of all, the model determines IoT device type according to several pre-assembled samples. The next steps involve looking for anomalies and categorizing different infiltration types by using four testing scenarios. [17] discovered that the decision trees model, which is a tree-based classifier, was the most effective of the models in terms of the F-1 indicator (about 98% accuracy) as a result of all experiments. However, it was found that the volume of data given to the test model was not sufficiently actual or unbiased, leading to an excessive level of accuracy in the model. Hence, researchers plan to use an upgraded version of the algorithm for quicker real-time performance.

On the contrary, the scientific work of [20] used a deeper comparative analysis based on various indicators of mean square error (MSE), root means square error (RMSE), mean absolute error (MAE), coefficient of determination (R2), etc. The article highlights the challenges faced by data center networks because of services' rapid expansion, which causes the ephemeral nature of traffic. In practical terms, data centers are a crucial architecture that cannot be left unprotected from external threats and hacker interventions. That is why the study attempts to apply a variety of test models, such as tree-based models and support vector machines (SVM). In order to simplify the process of model creation, the authors have used an open-source analysis and visualization tool which is named "Orange Data Mining". The results showed that the boosting algorithms, such as adaptive boosting (AB) and extreme gradient boosting (XGBoost), outperformed other machine learning algorithms in predicting network traffic, with AB with linear and exponential loss functions and XGBoost showing the best performance. According to [20], their predictive intrusion models could be implemented in other fields in the future.

[21] combined ML and deep learning into a hybrid approach to handling complex and large-scale datasets. Researchers combined the performance of an artificial neural network (ANN) and support vector machine (SVM). Also, they provided a complex feature selection method by wrapper technique. The study used the NSL-KDD dataset of malicious and benign network traffic with around 25000 test instances. The research outcome showed that ANN detection with 97% of success rate overperformed the SVM-based part of the model.

[22] worked with the same dataset. However, their experimental results were based only on tree-based models, which led to the highest result of 80% of final accuracy. They also weakly disclosed the topic of the main feature selection because [22] developed only an attribute discretization feature selection system which is named the chi-squared test.

[23] used the CSE-CIC-IDS2018 Amazon dataset that was captured during ten days of attacks or intrusions. There were several types of at-

tacks, like denial of service (DoS), brute force, botnet, etc. In this case, the authors performed four layered feature selections. Firstly, Pearson correlation calculates the degree of correlation between any two variables. Secondly, the chi-square test is used as a statistical method for assessing how closely expected, and actual results match each other. Thirdly, there is an additional technique of random forest for calculating relevance based on permuted out-of-bag samples. Fourthly, the deep learning layer adds access to make identification of optimal feature subsets. Only four features are sufficient to build efficient ML models that are comparable to those trained on all features, which can lead to simpler, more explainable, and more scalable models with reduced training time by 10% to 50%.

2.2. Deep learning models

Experiments involving deep learning techniques identify network traffic using the most unconventional techniques. According to the latest research by [24], common intrusion detection systems are useless or ineffective due to large amounts of data and a range of attack kinds. In this regard, authors propose to reshape traffic information into the QR code format for hybrid visual training of neural network models. MobileNetV2 and ShuffleNet CNN models are used to build and train QR code pictures for large amounts of data with various classes. The training pictures Deep CNN models and features are retrieved, and the Harris Hawk Optimization (HHO) method is then used to choose the best features for classification. In comparison to CNN models, the suggested hybrid model HHO detects attack types with an accuracy rate of 95.89%, offering improved performance. In their upcoming research, they want to use several CNN models, optimization strategies, and grayscale data sets to increase accuracy.

[25] suggest using the VSR-NLMS adaptive intrusion prediction method in order to apply it in real-time for mobile networks and traffic. Because its sample rate fluctuates with the traffic, the experiment demonstrates that VSR-LSTM has a greater level of traffic forecast accuracy. Meanwhile, [26] and [27] summarized the cumulative outcomes of the use of several deep learning models in various datasets or circumstances where network traffic was penetrated by hackers. The researchers found that the most popular open-source datasets for analysis in this field are KDDCup 99, NSL-KDD, Kyoto, UNSW-NB15, WSN-DS, etc. Also, deep neural networks (DNN), convolutional neural networks (CNN), recurrent neural networks (RNN) and deep migration learning (DML) are the main cybersecurity intrusion detection models in deep learning.

2.3. Text vectorization approaches

It should be noted that in addition to the classification of network traffic, researchers also employ numerous techniques for content grouping and visualization. For example, [28] discussed text vectorization techniques used to convert textual input into a numerical representation for ML model training. Authors identify five main types of clustering approaches: hierarchical clustering, k-means, k-medoid, DBSCAN, and DENCLUE. Hierarchical clustering is the simplest kind of clustering, the essence of which is the sequential union of groups. However, the centroid-shaped clustering approach is applied in the context of K-means and K-medoid.

[29] combined BERT and TFIDF vectorization principles to make a natural language processing (NLP) model of Twitter users' posts. Authors indicate that TFIDF is a commonly used technique to evaluate the relevance of a word in a collection of documents. Although this combination could affect better performances, [29] set limits of threshold to preserve the grammatical regularities in each document intact.

[30] manually added a natural language toolkit (NLTK) to their COVID-19 scam detection messages model. The system comprises two phases: the Domain Checking Phase and the SMS Classification Phase. The former focuses on scrutinizing the authenticity of URLs in the SMS,

while the latter identifies smishing messages based on a set of selected features. Furthermore, authors suggested that different text clustering techniques could be developed for creating a signature for the search engine domain or text checking.

The literature on cybersecurity highlights its evolving nature due to the increasing complexity and sophistication of cyber threats, driven by technological advancements and global digitalization. SIEM systems have emerged as critical components in modern cybersecurity strategies, providing organizations with centralized platforms to collect, monitor and analyze security-related data from various sources. Studies often emphasize the importance of SIEM in enhancing threat detection and response capabilities, enabling proactive security measures and facilitating compliance with regulatory requirements [7,16]. However, challenges such as data overload, false positives and the need for skilled personnel to manage SIEM systems are commonly discussed. Additionally, research focuses on integrating advanced technologies like ML and AI with SIEM to improve its efficiency and accuracy in identifying and mitigating security threats. Many studies underscore the pivotal role of SIEM in cybersecurity frameworks while also addressing its limitations and the ongoing efforts to overcome them through innovation and integration with cutting-edge technologies.

Despite the extensive research on cybersecurity and the acknowledged importance of SIEM systems in enhancing organizational security postures, there exists a notable gap in the literature concerning the practical implementation and effectiveness of SIEM solutions in real-world scenarios. Many studies focus on the technical aspects and theoretical frameworks of SIEM but often lack empirical evidence or case studies demonstrating their actual impact on improving cybersecurity outcomes. Additionally, there is limited research exploring the challenges organizations face in deploying and maintaining SIEM systems, particularly in terms of cost-effectiveness, scalability and usability. Furthermore, the integration of emerging technologies like ML and AI with SIEM for advanced threat detection and response remains an area that requires more in-depth exploration and evaluation. Thus, there is a need for more comprehensive and practical research that bridges the gap between theory and practice, offering actionable insights for organizations to effectively leverage SIEM solutions in their cybersecurity strategies.

3. Methodology

This section serves as the blueprint for our investigative journey, outlining the systematic approach employed to achieve our research objectives. In this section, we detail the carefully crafted steps, techniques, and tools used to gather, process, and analyze data. It provides a clear roadmap for how we explore the research questions, test hypotheses, and derive meaningful insights. Our methodology encompasses data acquisition, preprocessing, model selection, and evaluation procedures, ensuring rigor and reproducibility in our study's findings.

3.1. System overview

The current project system consists of two primary models: text information clustering based on TF-IDF and K-means, RFC intrusion prediction. This system incorporates the powerful techniques of TF-IDF and K-means clustering as integral components for the initial phase of threat detection. TF-IDF enables us to effectively process and represent textual security logs, while K-means clustering helps in grouping these logs into meaningful clusters based on their semantic similarities. First of all, there is collected network data from a test virtual machine on which a SIEM system is installed to collect traffic and attempts of brute force attacks. Then, this data is preprocessed to find null or zero values. Also, it is combined with the CICIDS2017 dataset of web attacks [31] and brute force history penetrations. In the next step, brute-force textual information is clustered into two main clusters, which are information technologies-related words and incoherent data. At this stage, visualization of additional information is demonstrated in 3D, 2D, or

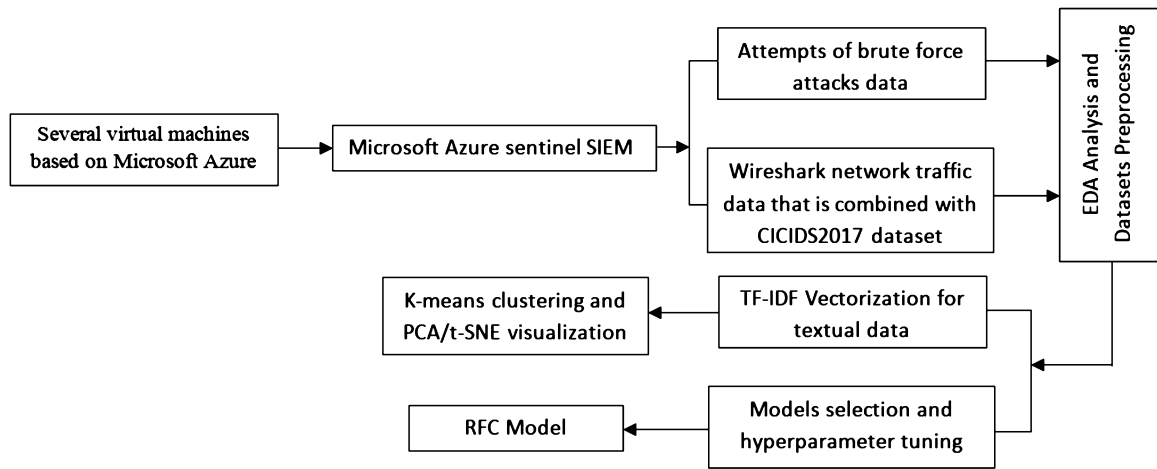


Fig. 1. Visual scheme of the system: This visual scheme illustrates the key components and processes within the system.

map-like plots. There are being implemented findings of features importance (found ten main features). Finally, the model is tuned using hyperparameter optimization, results of the trials are used to determine how well the model applies to a test dataset. Fig. 1 gives an overview of our model steps.

Our model begins with “Several virtual machines based on Microsoft Azure,” serving as the foundation for data collection. The security events and attempts of brute force attacks are monitored and logged within Microsoft Azure Sentinel SIEM. The data collected, which includes both “Attempts of brute force attacks data” and “Wireshark network traffic data combined with the CICIDS2017 dataset,” undergoes a crucial phase of analysis and preprocessing denoted as “EDA Analysis and datasets preprocessing.” During this stage, the data is cleansed, transformed, and prepared for subsequent analysis. The “TF-IDF vectorization for textual data” step involves the conversion of textual data into numerical vectors, making it suitable for machine learning. This vectorization process enables the system to work with textual features effectively. Moving forward, in the “Models section and hyperparameter tuning,” various machine learning models are trained and fine-tuned to optimize their performance. This step ensures that the models are well-prepared to handle the incoming data. The “K-means clustering and PCA/t-SNE visualization” phase employs clustering techniques for data segmentation and dimensionality reduction techniques for visual representation. These processes allow for a deeper understanding of data patterns and relationships. Finally, the “RFC model” emerges as the top-performing model, demonstrating its ability to effectively classify and interpret the data, making it a valuable asset in the system’s overall cybersecurity strategy.

3.2. Hosting VM and data acquisition

For this experiment, there were prepared several test virtual machines, such as honeypot devices. Honeypot is a cybersecurity mechanism designed to deceive and lure attackers into a controlled environment, where their actions and tactics can be monitored, analyzed, and studied without putting real systems or data at risk. Honeypots are an important tool in cybersecurity for understanding and defending against various forms of malicious activity, such as hacking, malware, and intrusion attempts. In addition, honeypots are intentionally set up to appear as attractive targets to cyber attackers. They mimic legitimate systems, applications, or services and are designed to lure malicious actors into interacting with them. Furthermore, honeypots play a critical role in cybersecurity by providing insights into the tactics and techniques used by malicious actors. They help organizations improve their security posture by allowing them to better understand threats, vulnerabilities, and potential attack vectors. However, deploying and managing

honeypots requires careful planning and expertise to ensure they are used effectively and do not inadvertently create security risks.

Data acquisition is the process of collecting and gathering data from various sources and making it available for analysis or storage. This process has been used in our study in order to collect data. The data acquisition environment, which refers to the tools, techniques, and infrastructure used to facilitate this process, has been used [32]. First of all, data acquisition begins with identifying and accessing relevant data sources (in our case from Honeypot, internet and virtual machines). Once data sources are identified, data collection methods are employed to retrieve data. Then, data is often noisy, incomplete, or inconsistent. Data preprocessing is performed to clean, transform, and format the data into a usable and structured format. This includes tasks like data cleaning, missing value imputation, and data normalization (see section 3.3). In many cases, data is acquired from multiple sources and needs to be integrated into a single dataset. In our case, CICIDS2017 [33,34] merged with the collected data in order to have one dataset.

An attractive target for intruders who want to gain unauthorized access to the device’s system or local network [35] is created. Honeypot, at the same time, simultaneously collects data on internet traffic into a single dataset in order to further conduct a qualitative analysis and compile a model by types of attacks, methods and tools of the attacking side [36]. Group of virtual machines was deployed on the basis of Microsoft Azure hosting. This service provides a free 30-day trial for all Microsoft account holders. Microsoft Azure also offers a wide range of services in the category of SIEM applications. The Azure logical group included: test virtual machines based on Windows 10 operating system, Azure Sentinel SIEM system for collecting internet traffic data, as well as a workspace for visualizing temporarily incoming information from the system logs. The self-written script was run on virtual machines that save the history of login attempts and its actions to check for brute force attacks. The collected information was converted into two new independent datasets. The first dataset is a combination of CICIDS2017 (Canadian Intrusion Detection Evaluation Dataset of Thursday attacks) with Wireshark traffic data of VMs that is used to create an intrusion detection model [37]. The second dataset is directly delivered from Azure SIEM that is used for clustering analysis.

3.3. Datasets preprocessing

Azure SIEM dataset consists of various columns: destinationhost_CF and sourcehost_CF (represent the name of our VM), label_CF (IP address of intruders), county_CF (country of intruders), state_CF (state or region of intruders), latitude_CF (north or south of the equator measuring), longitude_CF (east or west of the prime meridian measuring), timestamp_CF (timestamp of intrusion), username_CF (usernames

Table 1

Dataset example: This example displays the country_CF and sourcehost_CF columns from the Azure SIEM dataset. The sourcehost_CF column contains the source IP addresses associated with external entities, such as intruders or attackers, providing valuable insight into the origins of security incidents. These columns play a crucial role in identifying and analyzing potential threats within the Azure environment.

	timestamp_CF	county_CF	state_CF	sourcehost_CF	username_CF	longitude_CF	latitude_CF
0	2023-02-06T09:07:25Z	Belgium	Brussels	181.214.218.45	RECEPTION2	4.35609	50.84439
1	2023-02-06T09:07:23Z	India	Maharashtra	14.98.168.90	for	73.01204	19.14248
2	2023-02-06T09:07:21Z	Netherlands	North Holland	181.214.206.71	ADMINISTRATOR	4.87326	52.37018
3	2023-02-06T09:07:20Z	India	Maharashtra	14.98.168.90	for	73.01204	19.14248
4	2023-02-06T09:07:19Z	Netherlands	North Holland	181.214.206.156	MED	4.87326	52.37018
...	...	...	...	...	...	...	...
15841	2023-02-05T09:22:14:21Z	Kazakhstan	NaN	46.34.195.4	aituadmin	76.95694	43.25226
15842	2023-02-05T09:22:45:01Z	Kazakhstan	NaN	46.34.195.4	aituadmin	76.95694	43.25226
15843	2023-02-05T09:22:54:41Z	United States	Washington	20.231.60.181	AZUREADMIN	-122.89166	47.03956
15844	2023-02-05T09:23:34:11Z	Singapore	NaN	20.195.57.230	ADMINUSER	103.85211	1.29041
15845	2023-02-05T09:23:38:25Z	Singapore	NaN	20.195.57.230	ADMINUSER	103.85211	1.29041

that intruders used for brute forcing). Columns destinationhost_CF and sourcehost_CF are not relevant or valuable for the analysis. Hence these columns are dropped from the data frame. Moreover, due to 3069 empty lines in the state_CF column, there was applied to handle empty data with replacement to "N/A". It is essential to handle missing data appropriately, as ignoring or deleting missing data could lead to biased or inaccurate results. Additionally, converting the date from ISO 8601 format to datetime format allowed for easier manipulation and analysis of the dates. ISO 8601 [38] is a standard format for representing dates and times in Microsoft Azure and it is often used in data interchange formats. However, the datetime format is a more versatile format for working with dates and times in Python.

TF-IDF, which stands for Term Frequency-Inverse Document Frequency, is a numerical statistic used in natural language processing and information retrieval to evaluate the importance of a term within a document relative to a collection of documents (or corpus). TF-IDF is widely used in various applications, including text mining, search engines, document classification, and information retrieval systems. TF-IDF has been used in our model to process data. TF-IDF is a valuable tool for understanding and quantifying the importance of terms in text data.

As regards combined Wireshark and CICIDS2017 log files, there were deleted blank records in every column, converted string characters to numerical with One-Hot Encoder and TF-IDF. In addition, using both One-Hot Encoding and TF-IDF in a ML pipeline, which has been proposed, can be particularly effective when dealing with text data that contains categorical features and textual content such as in our case. First, dataset, which includes both categorical features and text data, has been prepared by properly cleaning and formatting. To address class disparities, the undersampling process removed and balanced data in a proportion of 30% attacks and 70% benign data records. Table 1 shows some examples of the dataset tables. Identify which columns in our dataset are categorical and which contain text in the second steps. Then, One-Hot Encoding has been applied for categorical features. It converts categorical variables into binary vectors, making them suitable for ML algorithms in our model. Furthermore, TF-IDF vectorization has been applied for the text data in order to convert the textual content into numerical features. Finally, after One-Hot Encoding and TF-IDF vectorization, two sets of features have been created. One stands for the categorical and the second for the text data. At this stage, with the combined feature set of our data, ML model could be trained.

3.4. Features selection

There were added decision tree visualization and single-tree evaluation of features significance (DecisionTreeClassifier). To make the classifier's presentation easier at first, utilizing one tree. In addition to decision tree visualization and single-tree evaluation of feature sig-

nificance using the DecisionTreeClassifier, feature selection is a critical step in the model-building process. It plays a pivotal role in ensuring that the classifier is provided with the most relevant and informative features while discarding irrelevant or redundant ones. In our analysis, we employed a combination of methods to select the most important features for our DecisionTreeClassifier.

Recursive Feature Elimination (RFE) [39] has been used. It is a technique that recursively fits the classifier and ranks features by their importance. It starts with all features and iteratively removes the least important ones based on their ranking. This process continues until the desired number of features is reached. We used RFE to select a subset of features that demonstrated the most discriminative power for our model. Besides using DecisionTreeClassifier for single-tree evaluation, we also leveraged ensemble methods such as Random Forest or Gradient Boosting to assess feature importance. Ensemble models aggregate the opinions of multiple trees, offering a more robust assessment of feature relevance. Furthermore, we conducted correlation analysis to identify pairs of features that were highly correlated. Highly correlated features can introduce multicollinearity, which can affect the stability of decision tree models. We either removed one of the correlated features or applied feature engineering to combine them into a single, more informative feature. Finally, cross-validation techniques have been applied to assess the stability and consistency of feature selection across different folds of the dataset. This helped us verify that the selected features consistently contributed to model performance.

High cross-validation scores, even with five leaves, appear suspiciously excellent. The decision to carefully examine high cross-validation scores, even with a small number of leaves (five in our case), is rooted in the possibility of overfitting. While high scores can be indicative of a well-performing model, they can also be a sign that the model has learned noise in the data, especially if the feature selection process is not rigorous. Therefore, our feature selection process was designed to strike a balance between achieving strong model performance and maintaining model generalization on unseen data, ensuring that our classifier is robust and interpretable.

3.5. Models selection with performance tuning

The quality of the responses of models was compared using the following metrics: accuracy, precision, recall, and F1-measure [40]. Accuracy measures the proportion of correctly classified instances out of the total instances in the dataset. It is a measure of overall model correctness:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \quad (1)$$

Precision measures the proportion of true positive predictions (correctly identified positive instances) out of all instances predicted as

Table 2

Results of the quality assessment of ten classifiers: KNN; SVM; CART; RF; AdaBoost; LR; NB; LDA; QDA; MLP.

Models	Accuracy	Precesion	Recall	F1-measure	Time(s)
KNN	0.971	0.942	0.961	0.960	4.57
SVM	0.705	0.669	0.036	0.602	176.04
CART	0.975	0.973	0.946	0.969	1.53
RF	0.971	0.978	0.943	0.970	1.14
AdaBoost	0.978	0.962	0.965	0.973	23.40
LR	0.955	0.939	0.914	0.963	15.80
NB	0.722	0.520	0.956	0.754	0.47
LDA	0.939	0.921	0.872	0.941	2.23
QDA	0.872	0.978	0.597	0.949	1.28
MLP	0.904	0.921	0.912	0.776	93.83

positive. It is a measure of how many of the positive predictions are actually correct. It is calculated as follows:

$$Precision = \frac{True\ Positives}{True\ Positives + False\ Positives} \quad (2)$$

Recall measures the proportion of true positive predictions out of all actual positive instances in the dataset. It is a measure of how well the model captures positive instances. Recall formula is as follows:

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Negatives} \quad (3)$$

F1-measure is the harmonic mean of precision and recall. It provides a balanced measure of a model's performance, considering both false positives and false negatives. The formula of F1-measure is:

$$F1 - measure = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (4)$$

The acquired quality metric values are displayed in the table below (Table 2), averaged over the outcomes of 5 cross-validation repetitions. Expectedly, the models (algorithms) KNN, CART, RFC, AdaBoost, and LR showed the best results. The adoption of the random forest classifier (RFC) model to solve the issue makes sense when the shortest execution time is taken into consideration. Also, there was a selection of hyperparameters for this model using the Grid-SearchCV (Grid Search Cross-Validation) technique.

Grid-SearchCV [41] is a technique used in ML to systematically search for the optimal combination of hyperparameters for a model. After running Grid-SearchCV, a set of results, which provide insights into the best hyperparameters, was obtained. These results often include various performance metrics (accuracy, precision, recall, F1-measure).

The Silhouette Method and the Elbow Method have been used in our study. They are two common techniques used in unsupervised ML, particularly for clustering analysis, to help determine the optimal number of clusters (k) in a dataset [42]. Each method provides insights into the appropriate choice of k based on different aspects of the data's structure. The Silhouette Method measures how similar each data point in one cluster is to the other data points in the same cluster compared to the nearest neighboring cluster. It assigns a silhouette score to each data point, and the average silhouette score across all data points is used to determine the optimal number of clusters. High Silhouette score (close to 1) indicates that the data point is well-matched to its own cluster. However, low Silhouette score (close to -1) means that the data point is poorly matched to its own cluster and well-matched to neighboring clusters, suggesting that k may be too large. In the other side, the Elbow Method focuses on the variance within the data as a function of the number of clusters. It looks for the "elbow point" on a plot of the number of clusters (k) versus the corresponding within-cluster sum of squares (WCSS). The WCSS measures the total variance of data points from their respective cluster centroids. The idea is that as k increases, WCSS decreases, but at some point, the decrease in WCSS becomes less pronounced, forming an "elbow" shape on the plot.

In the case of the text clusterization model, Silhouette and Elbow methods showed incorrect or impractical numbers of clusters for analy-

sis (see Fig. 2). That is why it was manually found that $k = 2$ is the best amount of clusters (useless and valuable data). K-means assigned each data point to k numbers of clusters.

4. Results and discussion

Results of our study showed that the feature importance algorithm based on the Decision Tree Classifier found the most valuable features on combined CICIDS2017 dataset. The feature space was narrowed down to the union of 10 features after the least significant features were eliminated: "Average Packet Size" (average length of TCP or IP packet data), "Flow Bytes/s" (data flow rate), "Max Packet Length" (maximum packet length), etc. Results of features selection are presented in Fig. 3.

Reducing the feature space to 10 features can be motivated by several reasons. In our case, it has been reduced in order to reduce dimensionality. Dealing with high-dimensional data, reducing the feature space to a lower dimension could help us mitigate the "curse of dimensionality." High-dimensional data often requires more data to model effectively, and it can lead to increased computational complexity and overfitting. By selecting a subset of the most relevant features, we managed to simplify the problem and improve model performance. Furthermore, feature selection is the process of choosing the most informative and relevant features while discarding less important ones. This could lead to more efficient and interpretable models. Selecting 10 features was based on domain knowledge, feature importance analysis and experimentation. In the other side, working with a reduced feature space made data visualization more manageable. Finally, a smaller feature space may help prevent overfitting, where a model becomes too specialized to the training data and performs poorly on new, unseen data. By reducing our feature space result in models generalized better.

Number of trees in the forest (n estimators), the minimum number of objects in one tree leaf (min samples leaf), the maximum tree depth (max depth), and the maximum number of features for one tree (max features) were selected among the model's adequate hyperparameters. The value of the F1-measure was used to determine the level of quasi-optimality of the model parameters. Fig. 4 gives Grid-SearchCV results of hyperparameter tuning.

The configured and trained "RandomForestClassifier" model demonstrated strong performance on the test sample, achieving a recall score of 0.96 and an F1 score of 0.97. This suggests that the model is highly effective at identifying true positives, with only a small number of false negatives. The high recall score indicates that the model has a low rate of missing relevant cases, which is critical in applications such as medical diagnosis or fraud detection. Additionally, the high F1 score indicates a good balance between precision and recall, meaning that the model is able to achieve high accuracy while also minimizing false positives. Overall, the results of the model evaluation suggest that the Random Forest Classifier is a highly effective algorithm for the given classification task.

K-means text clustering after its vectorization by TF-IDF showed positive results on the indicator of two clusters, dividing the information into IT-related data and useless data. The IT-related data includes various information like "administrator", "VM", "account", and "computer". These are typical device names that ordinary network users give to their devices. This shows that it is necessary to develop your personal information security literacy and not call devices by their familiar names. An example of visualization of two primary clusters is shown in Fig. 5.

5. Conclusion

This research paper has delved into the dynamic intersection of SIEM systems with ML algorithms, shedding light on the transformative potential of this synergy in the realm of cybersecurity. Throughout this study, we have unveiled a series of critical findings and key takeaways

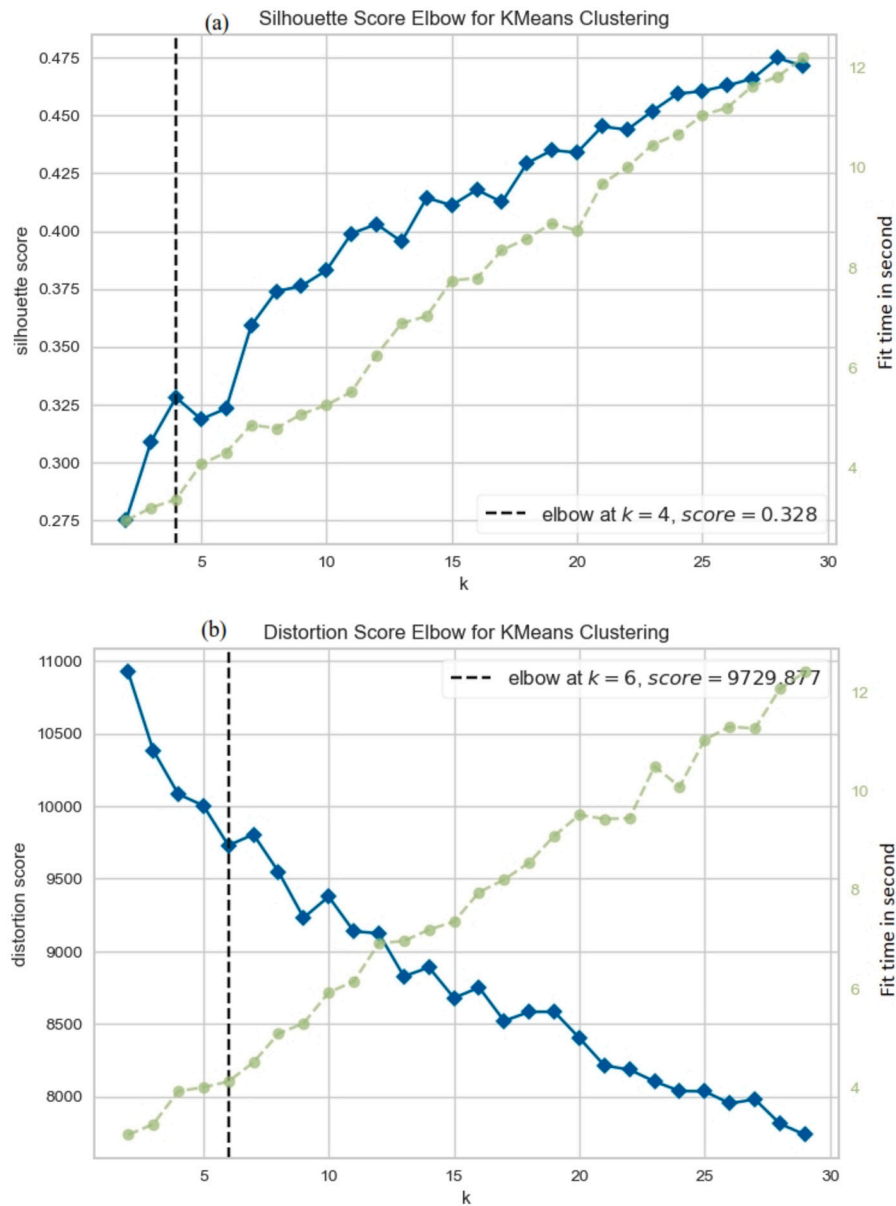


Fig. 2. Silhouette and Elbow methods: This figure illustrates the results of the Silhouette and Elbow methods used to evaluate the clustering performance. The Silhouette analysis (a): assesses the quality of cluster assignments, with higher values indicating better separation between clusters. The Elbow analysis (b): helps determine the optimal number of clusters by identifying the “elbow point” where the rate of variance reduction levels off.

that underscore the profound impact and promising future of SIEM powered by ML. The integration of ML algorithms into SIEM has ushered in a new era of proactive and adaptive security. These algorithms, capable of processing vast and diverse datasets with human-like precision, have significantly enhanced the detection, analysis, and response capabilities of SIEM systems. Their ability to identify subtle anomalies, predict emerging threats, and automate incident response tasks has become a game-changer in the fight against cyber adversaries.

Our exploration has revealed the versatility of machine learning-driven SIEM across various domains, including network intrusion detection, user behavior analysis, and anomaly detection. This adaptability empowers organizations to tailor their cybersecurity strategies to specific threats and vulnerabilities, offering a level of customization and accuracy previously unattainable. However, it is essential to acknowledge the challenges inherent in this evolving landscape. Successful implementation requires robust data management practices, careful model selection, and continuous monitoring and adaptation. Furthermore, the human-machine partnership remains a critical aspect of effective cy-

bersecurity, with human expertise complementing the capabilities of machine learning algorithms.

CRediT authorship contribution statement

Noyan Tendikov: Writing – original draft, Software, Investigation. **Leila Rzayeva:** Writing – original draft, Software, Methodology. **Bilal Saoud:** Writing – review & editing, Validation, Methodology. **Ibraheem Shayea:** Writing – review & editing, Validation, Methodology. **Marwan Hadri Azmi:** Writing – review & editing, Validation, Methodology. **Ali Myrzatay:** Writing – review & editing, Software, Investigation. **Mohammad Alnakhli:** Formal analysis, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

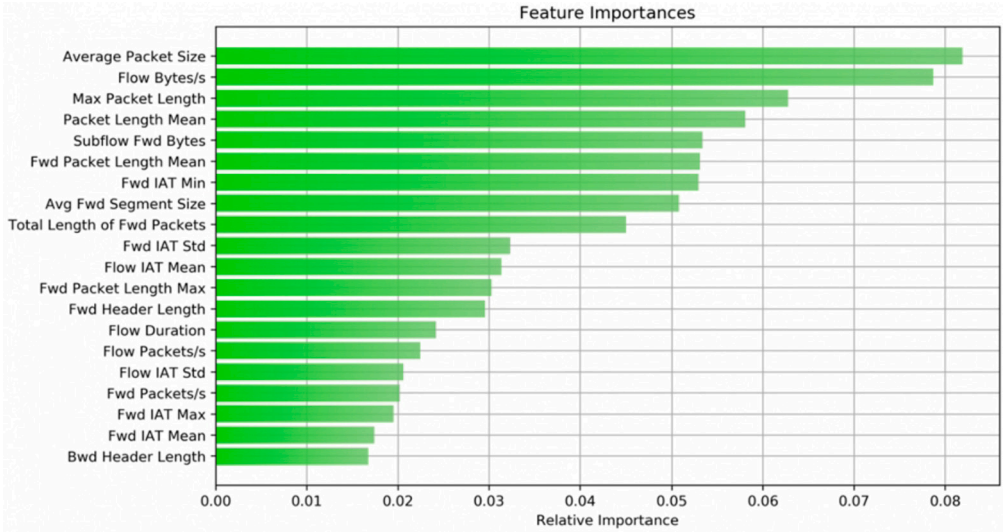


Fig. 3. Features selection results.

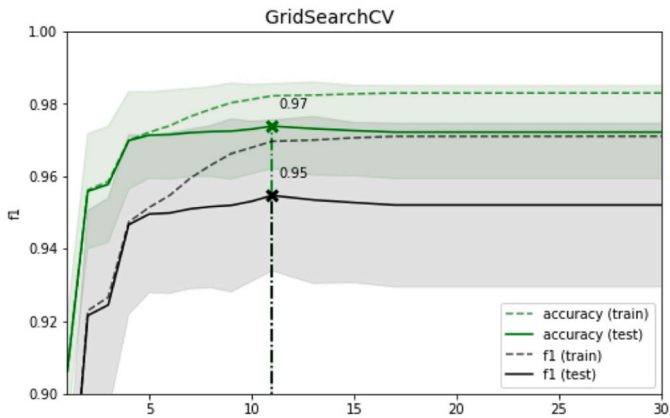


Fig. 4. Grid-SearchCV results of hyperparameter tuning.

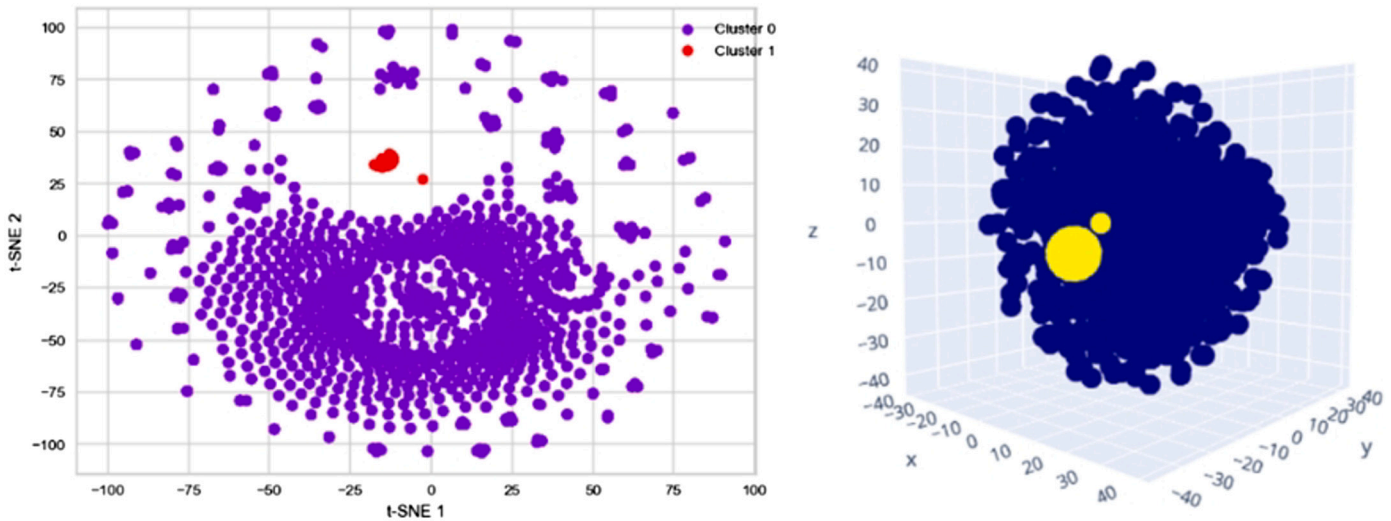


Fig. 5. Visualization of two primary clusters: This figure presents a visualization of the two primary clusters obtained through K-means text clustering after TF-IDF vectorization. The clusters effectively segregate the data into two distinct categories: “IT-related data” and “Useless data.” Notably, the “IT-related data” cluster includes common terms such as “administrator,” “VM,” “account,” and “computer,” which are indicative of typical device names assigned by network users. This underscores the importance of enhancing personal information security literacy and refraining from using familiar names for devices.

Data availability

Data will be made available on request.

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Appendix A. Table 3

Table 3

The definition of abbreviation.

Abbreviation	Definition
ICT	Information and Communications Technology
SIEM	Enter Security Information Event Management
IDS	Intrusion Detection Systems
AWS	Amazon Web Services
IT	Information Technology
DPL	Data Loss Prevention
AI	Artificial Intelligence
ML	Machine Learning
IoT	Internet of Things
MSE	Mean Square Error
RMSE	Root Means Square Error
MAE	Mean Absolute Error
R2	Coefficient of determination
XGBoost	Extreme Gradient Boosting
SVM	Support Vector Machines
AB	Adaptive Boosting
RF	Random Forest Classifier
KNN	K-Nearest Neighbors
CART	Classification and Regression Trees
RF	Random Forest
AdaBoost	Adaptive Boosting
LR	Logistic Regression
NB	Naive Bayes
LDA	Linear Discriminant Analysis
QDA	Quadratic Discriminant Analysis
MLP	Multilayer Perceptron
ANN	Artificial Neural Network
DoS	Denial of Service
HHO	Harris Hawk Optimization
DNN	Deep Neural Networks
CNN	Convolutional Neural Networks
RNN	Recurrent Neural Networks
DML	Deep Migration Learning
NLP	Natural Language Processing
NLTK	Natural Language Toolkit
CIDEDT	Canadian Intrusion Detection Evaluation Dataset of Thursday attacks
destinationhost_CF	Name of our VM
sourcehost_CF	Name of our VM
label_CF	IP address of intruders
county_CF	Country of intruders
state_CF	State or region of intruders
latitude_CF	North or south of the equator measuring
longitude_CF	East or west of the prime meridian measuring
timestamp_CF	Timestamp of intrusion
username_CF	Username that intruders used for brute forcing
RFE	Recursive Feature Elimination

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